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Description - The Power Monitor Test Aid Kit is a service tool that allows remote observation of the Power Monitor status LED's at the Operator Workspace during servicing. The tool may be used with Power Monitors that have RF Power Monitor Control assemblies 46-226748 or equivalent. The Kit is a modular assembly of four pieces, the ribbon cable, MUX assembly, four wire telephone cord, and the DEMUX assembly.

1- REQUIRED TOOLS

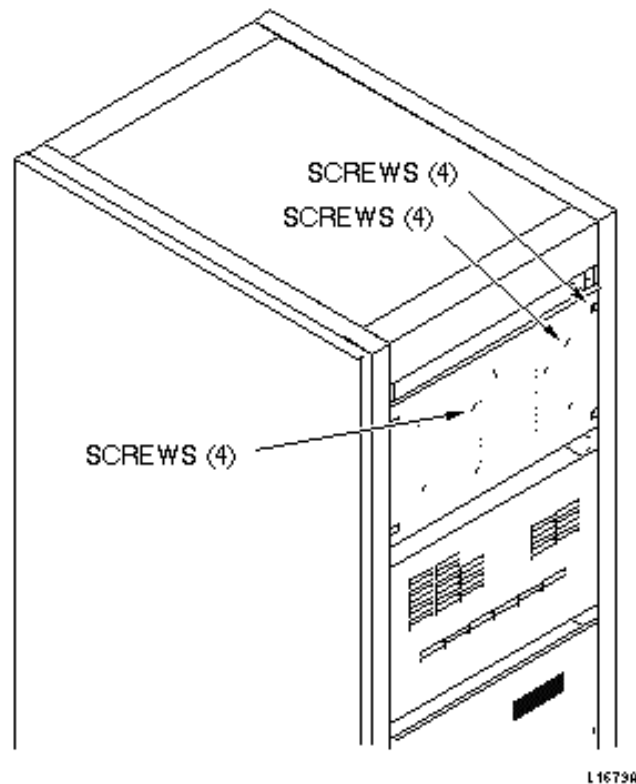
Refer to Table 1-1 for contents of Power Monitor Tool Kit 46-320557G1.

TABLE 1-1
REQUIRED TOOLS

Item	Description	Part Number	Quantity
1.	DEMUX Assembly	46-320484G1	1
2.	MUX Assembly	46-320485G1	1
3.	Phone Cable	46-320513P1	2
4.	Phone Adapter	46-320513P2	1
5.	Harness	46-320152G1	1
6.	Cable, Ribbon	46-320691P1	2

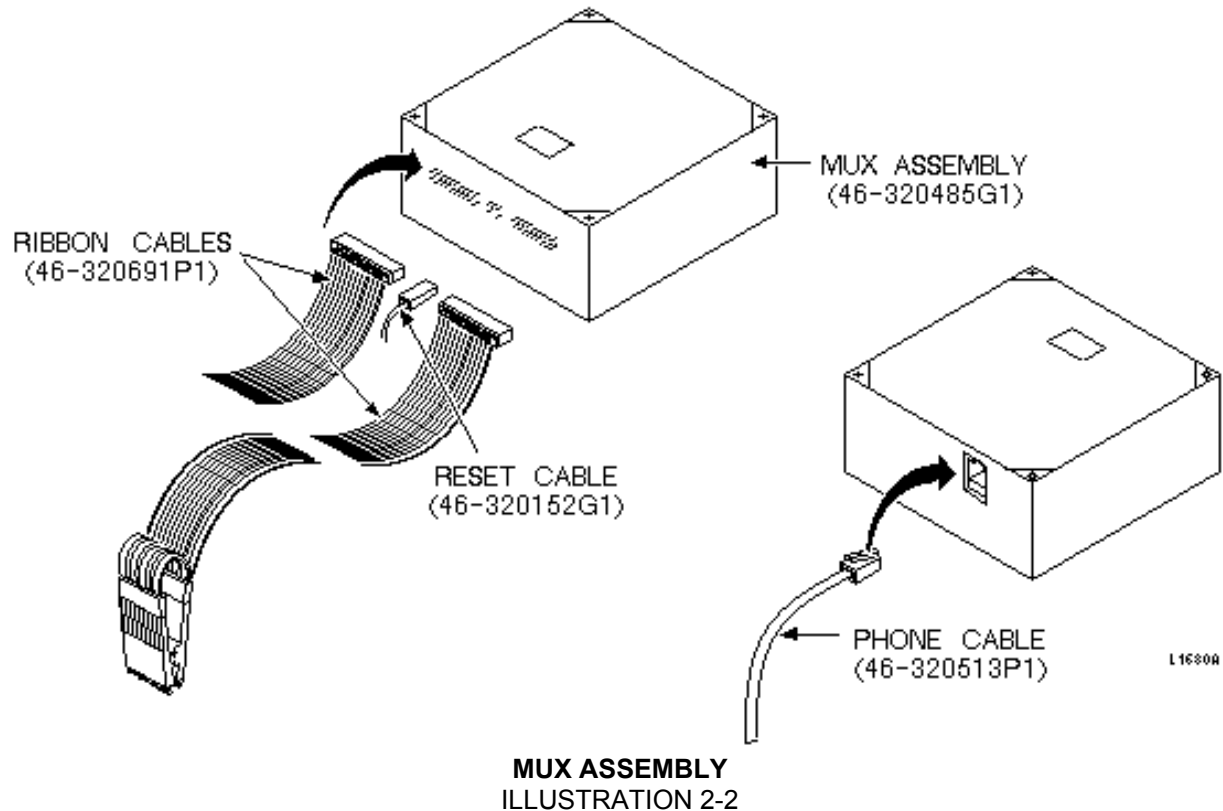
2- POWER MONITOR TOOL INSTALLATION

1. Remove twelve screws from front panel of Power Monitor. See Illustration 2-1.

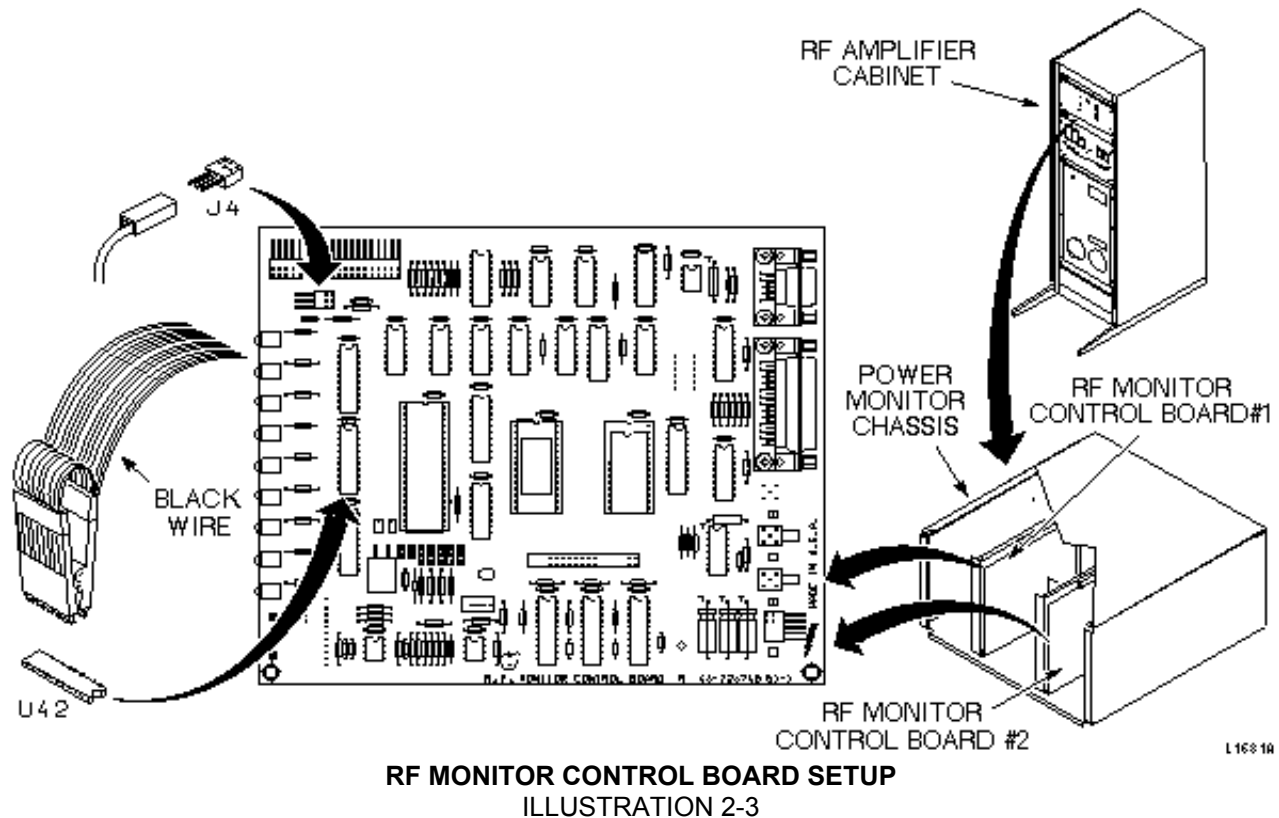


POWER MONITOR CHASSIS FRONT VIEW
ILLUSTRATION 2-1

2. Remove reset switch cables from J4 on each Power Monitor Control Board. Put front panel aside.
3. Connect the ribbon cables (46-320691P1) to monitor 1 (J1) and monitor 2 (J3) connectors on the Power Monitor MUX Assembly (46-320485G1). See Illustration 2-2.



4. Connect reset cable (46-320152G1) to J2 connector on MUX assembly.
5. Connect phone cable (46-320513P1) to J4 connector on MUX assembly.
6. Hang MUX assembly on side panel of Power Monitor chassis using the plastic hanger clip mounted on back of MUX assembly.
7. Connect Test Clip from MUX assembly, Monitor 1 to U42 on Power Monitor Control Board # 1. See Illustration 2-3.



8. Connect Test Clip from MUX assembly, Monitor 2 to U42 on Power Monitor Control Board #2.

Note

LED Light Status – Ensure both green LED's (Monitor 1 and Monitor 2) on the MUX assembly are lit after the test clips are connected to U42. This will indicate correct connection.

9. Connect the reset cables from the MUX assembly to J4 on each of the Power Monitor Control Boards.

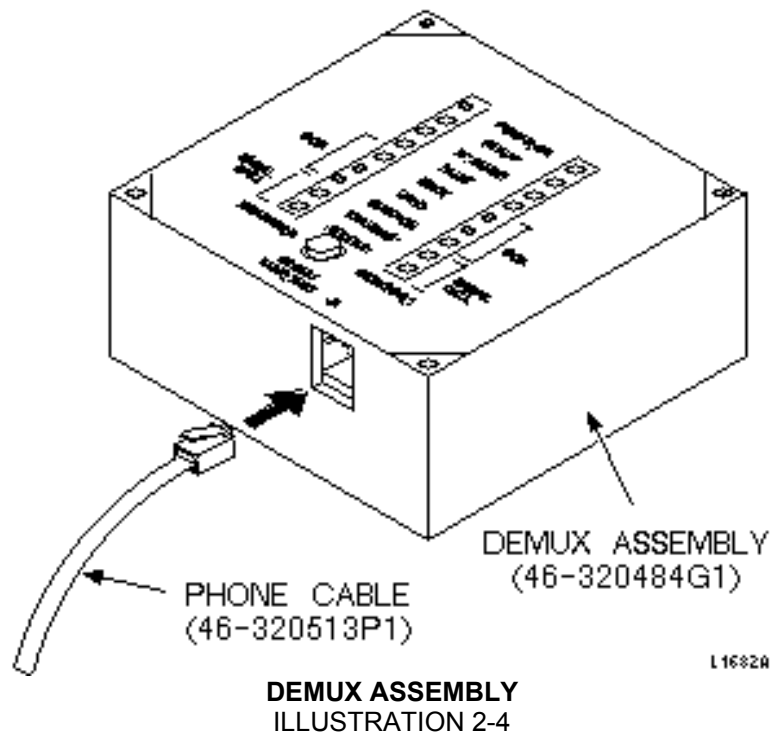
Note

Reset Cable Connection - It does not matter which reset cable is connected to which Power Monitor Control Board.

Note

Test Clip Connection - If test clip is improperly connected to U42 or another chip, tool has circuitry so that neither tool or control boards will be damaged.

10. Carefully lay the phone cable on the floor back to the Operator Console. If the distance to the Operator Console is greater than 25 ft., a second phone cable should be used. Attach the two phone cables together with phone coupler (46-320513P2).
11. Connect the phone plug to the DEMUX assembly at J1. See Illustration 2-4.



Note

"POWER ON" LED Light Status - If proper connections are made, the POWER ON LED's should now be lit.

12. Press the red "RESET/LAMP TEST" button on the DEMUX assembly. Ensure all the LED's light up to indicate they are functioning.
13. Secure the DEMUX assembly on top of the Operator Console.

Note

"RESET/LAMP TEST" Button - The red "RESET/LAMP TEST" button on the DEMUX assembly can be used to remotely reset the power monitor.

14. Run Power Monitor Functional Check.

3- RESTORATION CHECKLIST

1. Remove reset cables and test clips from RF Control Monitor Board.
2. Reconnect reset cables from Key Switch to Power Monitor Control Boards and install Power Monitor Front Panel onto Power Monitor chassis.
3. Return Power Monitor Tool items to their proper place in the kit container.

4- POWER MONITOR TOOL OVERVIEW

The Power Monitor Test Aid Kit is a service tool that allows remote observation of the Power Monitor status LED's at the system console during servicing. The tool may be used with Power Monitors that have RF Power Monitor Control assemblies 46-226748 or equivalent. The Kit is a modular assembly of four pieces, the ribbon cable, MUX assembly, four wire telephone cord, and the DEMUX assembly.

The MUX assembly receives inputs and power from the RF Power Monitor Control Board through the ribbon cable, samples and time division multiplexes each LED drive status, and sends the composite signal (along with power and ground) to the DEMUX assembly via the four wire telephone cord. The MUX will also reset the PM when a reset signal from the DEMUX assembly is received.

Four connections from the MUX assembly to the Power Monitor are necessary for complete operation. The 20 pin DIP clips (J1, J3) need to be attached to IC U42 and the reset connections (J2-1,2) to J4 on each Power Monitor. The assembly is powered on when the connections are completed to the Power Monitor.

The MUX is attached to the DEMUX via a standard four wire telephone cord. This allows the user to easily replace damaged cable with inexpensive off the shelf pre-connected telephone cord readily available at most retail hardware stores.

The DEMUX assembly receives the time division multiplexed signal, clock and synchronization pulse, and power from the MUX assembly across the four wire telephone cable. The Power Monitor LED information multiplexed on the signal line is decoded with the clock/synchronization signal and lights the corresponding LED on the face of the DEMUX assembly. The 18 LED display mimics the Power Monitor LED display. A switch "RESET/LAMP TEST" is provided to remotely reset the Power Monitor and perform a DEMUX lamp test.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 10, 1998	J. Saperstein	Initial conversion from Toolbook to Word formats.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document, removed obsolete manual reference, updated illustration numbers per style guide.